



LUIZ HENRIQUE PALUDO

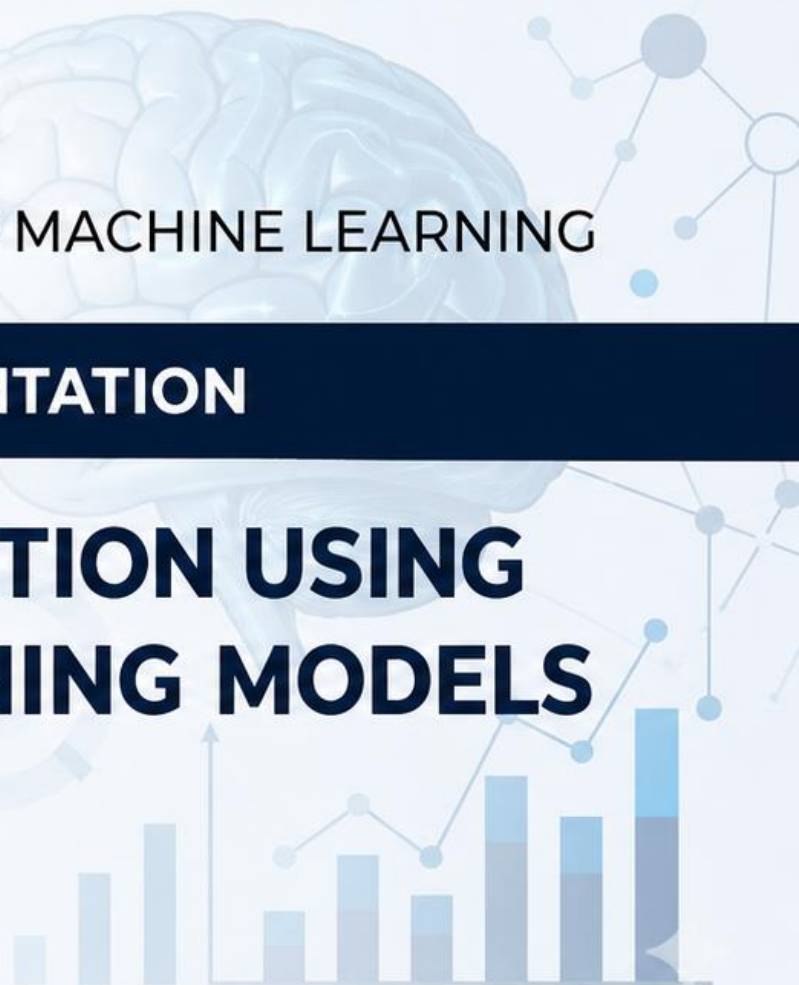
CLASS: INTRODUCTION TO MACHINE LEARNING

FINAL PROJECT PRESENTATION

STROKE PREDICTION USING MACHINE LEARNING MODELS



DATE: MAY 2026



Raw Dataset.

- 12 columns.
- With a total of 5110 entries.

Obs: Column numbers are 12 but the last one is 11, because in python lists starts in 0.

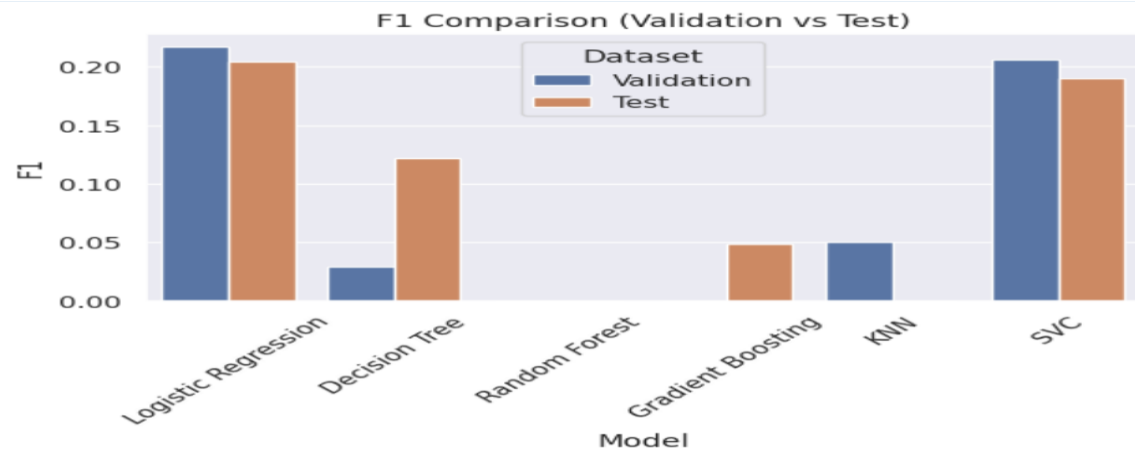
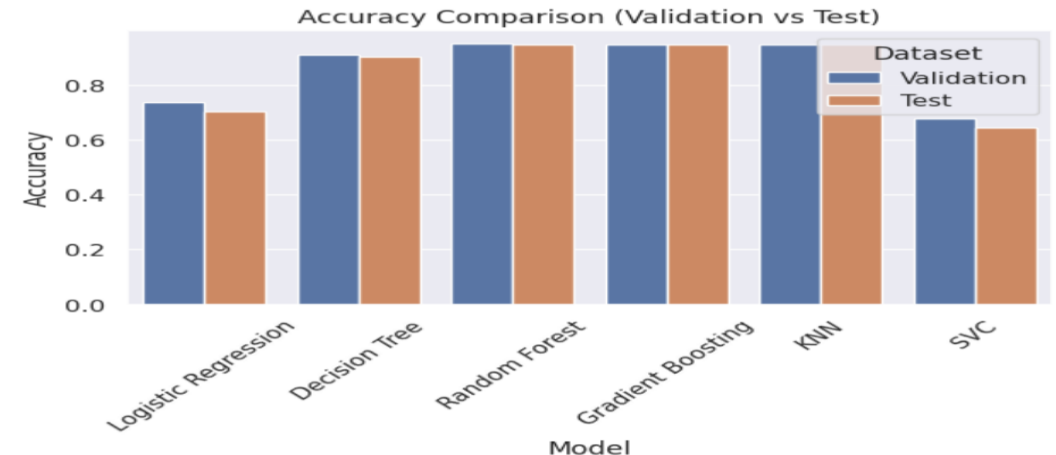
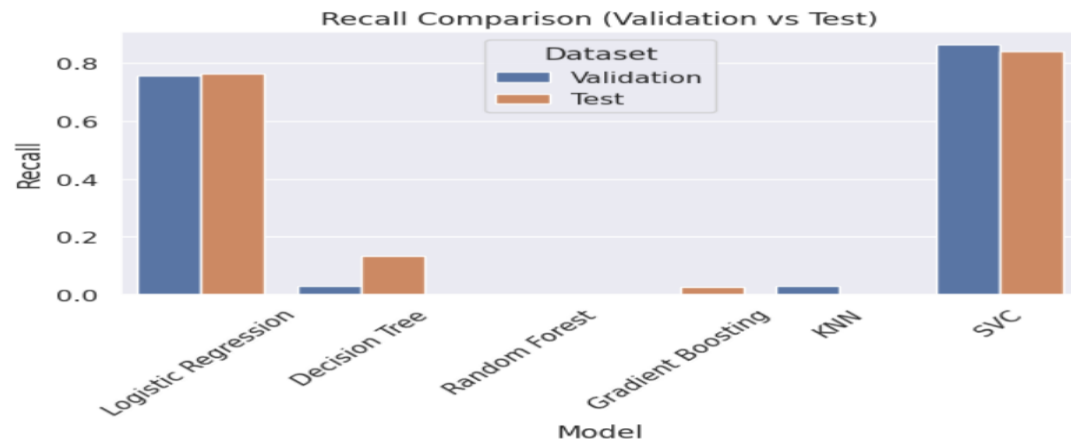
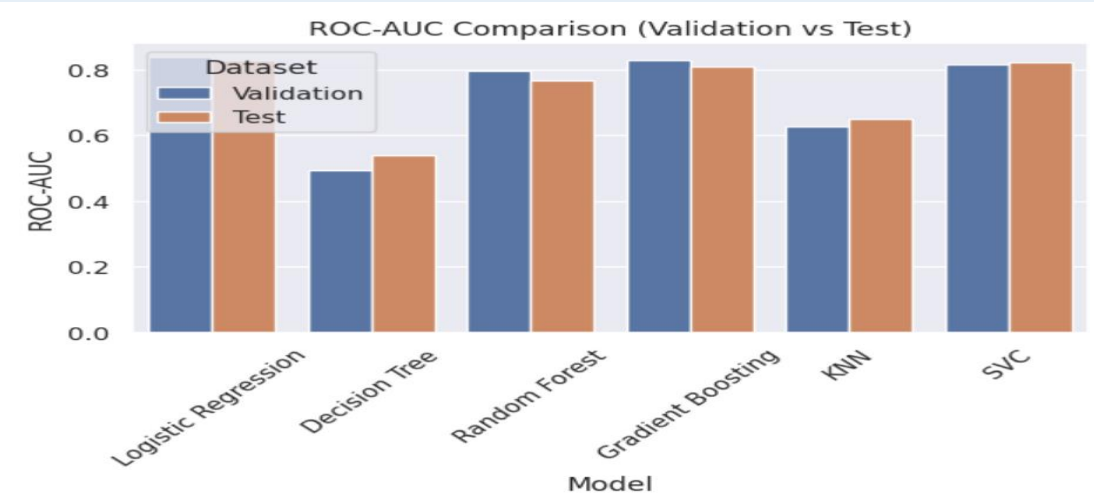
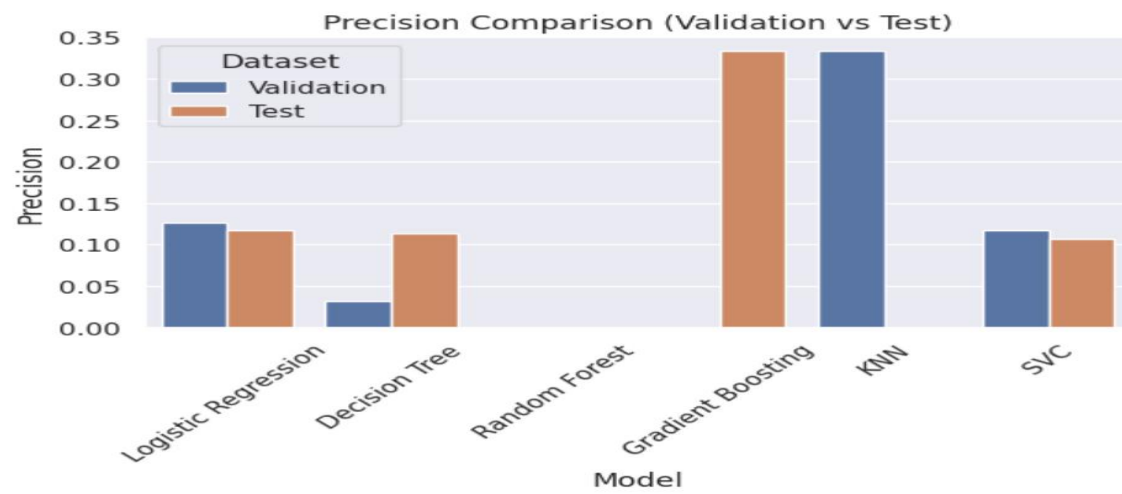
```
RangeIndex: 5110 entries, 0 to 5109
```

```
Data columns (total 12 columns):
```

#	Column	Non-Null Count	Dtype
---	-----	-----	-----
0	id	5110 non-null	int64
1	gender	5110 non-null	object
2	age	5110 non-null	float64
3	hypertension	5110 non-null	int64
4	heart_disease	5110 non-null	int64
5	ever_married	5110 non-null	object
6	work_type	5110 non-null	object
7	Residence_type	5110 non-null	object
8	avg_glucose_level	5110 non-null	float64
9	bmi	4909 non-null	float64
10	smoking_status	5110 non-null	object
11	stroke	5110 non-null	int64

```
dtypes: float64(3), int64(4), object(5)
```

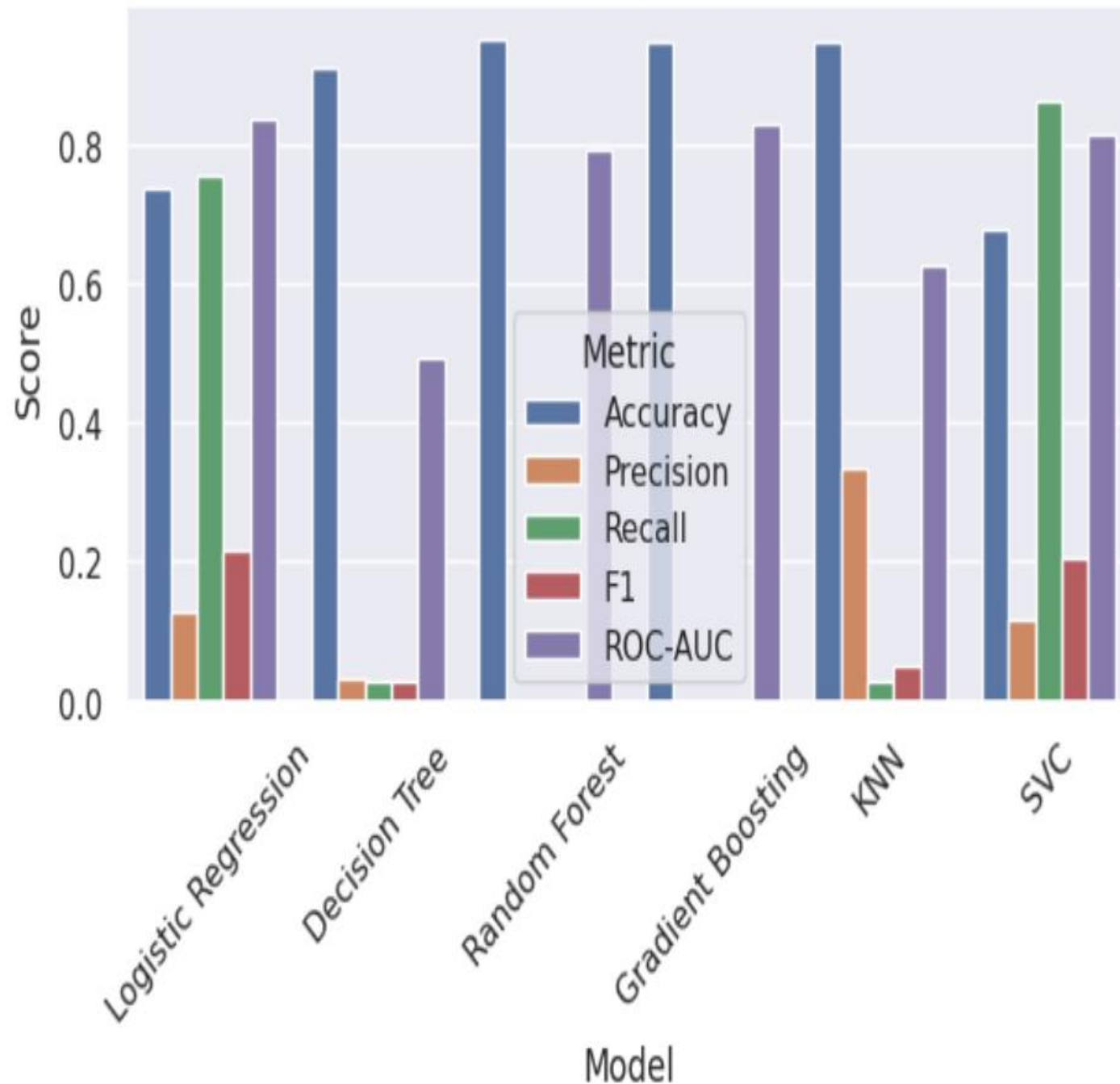
```
memory usage: 479.2+ KB
```



Validation VS Test

These graphs compare the performance of different machine learning models on validation and test datasets using metrics like precision, recall, accuracy, F1 score, and ROC-AUC to evaluate how well each model predicts stroke risk.

Model Performance Comparison (Validation Set)



Final Analyses.

Even though some models showed very high accuracy, they are not useful because they fail to detect positive cases. The best individual model is Logistic Regression because, it has a good balance between metrics, detects positive cases better and performs consistently.

Obs: SVC is also a strong model, especially for recall.

The ensemble model is the best overall because it combines multiple models and gives more reliable results.

Thank You